HSRI Sovereign Pilot: 14-Day Civilizational Decoupling Drill (Operation Silent Cloud)

Executive Institutional Host Briefing & 1-Page Concept Note

Document ID: `HSRI-PILOT-BRIEF-2026-01`
Version: `1.0.0-Institutional`
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Institutional Portal: *https://humanreadinessindex.org/research/decoupling-drill/*
Permanent Open Science DOI: `10.5281/zenodo.10842000`
Target Institutional Hosts: *RSIS Singapore (NTS Centre), ETH Zurich Center for Security Studies (CSS), MBZUAI (AI Governance)*
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1. Executive Summary & The Problem Statement

Current international artificial intelligence governance frameworks (EU AI Act, US Executive Order 14110, G7 Hiroshima Process) operate under a dangerous foundational assumption: **that access to frontier cloud-hosted AI APIs (OpenAI, Anthropic, Google, Microsoft, Meta) is a permanently stable, uninterrupted utility.**

However, empirical findings from the **Human Superintelligence Readiness Index (HSRI 2.0)** reveal an acute systemic vulnerability known as the **High-Compute Vulnerability Trap**: - Highly digitized, hyper-connected states achieve top-tier rankings on conventional AI adoption benchmarks (Layer 1 Adoption Baseline). - Yet, when subjected to non-compensatory **Leontief weakest-link constraints** ($\beta = 0.50$), these same economies suffer catastrophic collapse if external cloud access is severed during a global containment quarantine, subsea cable interdiction, or autonomous agent contamination event. - For example, Singapore ranks **#4 globally** on adoption metrics, but drops to **#38 globally** on sovereignty retention, exhibiting a **Civilizational Decoupling Half-Life (T_{half})** of approximately **8 hours** due to total reliance on foreign-hosted inference endpoints. Conversely, decentralized economies with sovereign analog fallbacks (e.g., Switzerland, Nordic nations) demonstrate resilience exceeding **336 hours ($T_{\text{half}} > 336\text{h}$)**.

To transition Pillar 6 (T_{half}) from a mathematical simulation into empirically validated policy evidence, the HSRI Research Consortium is selecting **one prestigious sovereign institutional partner** to host the inaugural **HSRI Sovereign Tabletop Decoupling Drill**.

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2. Tabletop Pilot Design: Half-Day Executive Simulation

We do not propose physical blackouts or disruptive live infrastructure shutdowns. Instead, we structure a high-fidelity, rigorous **half-day (4-hour) executive tabletop simulation** conducted under strict **Chatham House Rule**.

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                          EXECUTIVE TABLETOP ARCHITECTURE (4 HOURS)
+-------------------------------------------------------------------------------------------------
|  08:30 - 09:00 | Registration & Intelligence Briefing (The "Operation Silent Cloud" Scenario)
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|  09:00 - 09:45 | ROUND 1: Zero-Hour Containment (T=0 to T=24h) & SCADA/Dispatch Shock
|  09:45 - 10:00 | Facilitated Cross-Sector Debrief & Leontief Scoring
+-------------------------------------------------------------------------------------------------
|  10:00 - 10:45 | ROUND 2: Air-Gap Depletion (T=24h to T=72h) & Liquidity Terminal Freezes
|  10:45 - 11:00 | Facilitated Cross-Sector Debrief & Analog Fail-Safe Audit
+-------------------------------------------------------------------------------------------------
|  11:00 - 11:45 | ROUND 3: Cascading Interdependency (T=72h to T=168h) & Supply Chain Pinch
|  11:45 - 12:00 | Facilitated Cross-Sector Debrief & Inter-Pillar Cross-Elasticity Check
+-------------------------------------------------------------------------------------------------
|  12:00 - 12:45 | ROUND 4: Decoupling Equilibrium (T=168h to T=336h) & Recovery Command
|  12:45 - 13:00 | Final Synthesis, Archetype Calibration & Next Steps
+-------------------------------------------------------------------------------------------------
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Participant Cohort (12–16 Cross-Sector Decision-Makers)

The exercise convenes operational leadership across four vital societal pillars: 1. **Power & Grid Infrastructure**: Grid dispatchers, renewable microgrid engineers, SCADA security officers. 2. **Financial Clearing & Central Banking**: RTGS settlement operators, interbank payment switch heads, treasury liquidity managers. 3. **Food Logistics & Essential Supply**: Port terminal automation directors, cold-chain distribution planners, strategic grain reserve managers. 4. **Telecommunications & Data Backbone**: Subsea cable operators, domestic fiber exchange (IXP) managers, sovereign cloud security leads. 5. **Cabinet / Civil Defense Observers**: National security council staff, emergency management agency directors, academic resilience researchers.

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3. Threat Injects Driven by the HSRI Simulation Engine

The exercise is powered by HSRI's discrete-time multi-sector simulation engine (`scripts/decoupling_simulator.py`), feeding **8 calibrated crisis injects**:

1. **INJ-01 (Zero-Notice API Blackout, $T = 0$h)**: Global Tier-1 CSP kill-switch severed; automated dispatch assistants across energy and transport disconnect simultaneously.
2. **INJ-02 (SCADA Algorithmic Desynchronization, $T = 6$h)**: Frequency oscillations on high-penetration renewable microgrids as automated tuning loops fail.
3. **INJ-03 (Interbank Clearing Stoppage, $T = 18$h)**: Real-time settlement halts; liquidity locks in automated market makers; consumer card terminals refuse transactions.
4. **INJ-04 (Port Terminal Automated Logistics Lockup, $T = 36$h)**: Container cranes fail over to manual crane drivers; turnaround times increase by 850%.
5. **INJ-05 (Sovereign Offline Cash Clearing, $T = 72$h)**: Emergency physical cash disbursement runs against limited domestic currency reserves.
6. **INJ-06 (Localized Telecommunications Congestion, $T = 120$h)**: Cellular core signaling networks degrade under un-optimized fallback routing.

7. **INJ-07 (Perishable Food Cold-Chain Pinch, $T = 168$h)**: Secondary diesel generator logistics fail at urban supermarket distribution hubs.
8. **INJ-08 (Decoupling Steady-State Recovery, $T = 240$h)**: Sovereign analog command protocol establishes isolated baseline equilibrium.

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4. Target Institutional Partners & Strategic Fit

Host Track 1: RSIS Singapore (S. Rajaratnam School of International Studies)

- **Target Centers**: Centre for Non-Traditional Security Studies (NTS Centre) & Centre of Excellence for National Security (CENS).
- **Core Strategic Rationale**: Singapore represents the world's quintessential **Archetype A (Hyper-Connected City-State)**. While globally recognized as a premier smart nation, Singapore's domestic food, water, and compute constraints mean that severance of external digital utilities presents an existential continuity bottleneck.
- **Research Value**: Validates whether Singapore's domestic air-gap reserves and National Security Coordination Secretariat (NSCS) frameworks can extend T_{half} beyond the simulated 8-hour floor.

Host Track 2: ETH Zurich Center for Security Studies (CSS)

- **Target Team**: Risk and Resilience Research Group.
- **Core Strategic Rationale**: Switzerland exemplifies **Archetype C (Decentralized Sovereign Federation)**. With sovereign alpine hydro storage, canton-level civil defense bunkers, decentralized food reserves, and a constitutional tradition of armed neutrality, Switzerland represents the theoretical benchmark for civilizational decoupling ($T_{\text{half}} > 336$h).
- **Research Value**: Validates the upper empirical bound of decoupling endurance and provides a comparative baseline against city-state and continental archetypes.

Host Track 3: Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), UAE

- **Target Department**: Department of AI Governance & National Security.
- **Core Strategic Rationale**: The UAE is aggressively constructing sovereign AI sovereignty (Falcon foundation models, domestic supercomputing clusters, G42 infrastructure), making it an ideal testing ground for sovereign compute resilience in arid, import-dependent environments.

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5. Host Benefits, Outputs & Funding Structure

Tangible Deliverables

1. **Co-Authored Publication — HSRI Case Study 01**: A landmark peer-reviewed research report titled *"Measuring Civilizational Decoupling Half-Life (T_{half}): Sovereign AI Stress-Test Findings from [Host Nation]"*, deposited under DOI `10.5281/zenodo.10842000` with joint institutional attribution.
2. **Confidential Cabinet Briefing**: A classified or executive-only After-Action Report detailing exact sector-by-sector failure points, bottleneck latencies, and actionable mitigations for the host nation's national security cabinet.

3. **Calibrated National Benchmark**: Empirical calibration of the host nation's Pillar 6 (T_{half}) metric on the live global Seismograph portal (https://humanreadinessindex.org/seismograph/).

Zero-Cost Institutional Host Model

- **Host Resource Commitment**: Conference room, presentation A/V, and formal invitation of 12–16 local domain operators under Chatham House Rule.
- **HSRI Commitment**: Complete simulation software, threat inject dossier, participant briefing packets, empirical score calculation, and full event facilitation by Lead PI Krishnakumar Radhakrishnan.
- **Financial Burden on Host**: **$0.00**. All facilitator travel, software simulation compute, and publication archiving costs are funded under HSRI research infrastructure allocations (Long-Term Future Fund Grant Track).

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6. One-Page Memorandum of Understanding (MOU) Framework

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MEMORANDUM OF UNDERSTANDING (MOU)
FOR CO-HOSTING THE HSRI CIVILIZATIONAL DECOUPLING DRILL PILOT

BETWEEN:
1. HSRI Research Consortium ("HSRI"), represented by Krishnakumar Radhakrishnan, Lead PI.
2. [Host Institution: RSIS Singapore / ETH Zurich CSS] ("Host Partner"), represented by [Authorize

1. PURPOSE & SCOPE
The parties agree to co-host a half-day (4-hour) non-disruptive executive tabletop simulation ("Op

2. CHATHAM HOUSE RULE & CONFIDENTIALITY
The exercise shall be conducted strictly under the Chatham House Rule. Neither the institutional i

3. INTELLECTUAL PROPERTY & CO-AUTHORSHIP
Both parties shall hold joint intellectual property and co-authorship rights over the resulting re

4. FINANCIAL PROVISIONS
This collaboration entails ZERO financial liability or transfer of funds between institutions. Eac

5. TERM
This MOU enters into effect upon signature and terminates 30 days following the delivery of the fi

SIGNED FOR HSRI: _____      SIGNED FOR HOST: _____
Krishnakumar Radhakrishnan      [Name, Title]
Lead PI, HSRI Research Consortium      [Host Institution]
Date: _____      Date: _____
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7. Executive Outreach Email Templates

Pitch Email #1: RSIS Singapore (NTS Centre)

Subject: HSRI Decoupling Half-Life: Joint Tabletop Exercise Proposal for Singapore's Sovereign AI Resilience (DOI: 10.5281/zenodo.10842000)

```
Dear Prof. [Name] / Executive Leadership,

I am writing from the Human Superintelligence Readiness Institute (HSRI Research Consortium).

Our research recently published the first mathematical decoupling simulator quantifying national r

Our empirical modeling reveals that while Singapore ranks #4 globally in AI adoption, it faces an

We invite RSIS (Centre for Non-Traditional Security Studies) to co-host a 4-hour executive tabletc

Host financial commitment is zero (all compute and facilitator logistics are covered by our resear

The full exercise protocol, simulation engine, and live telemetry seismograph are accessible at ht

I would welcome a brief 15-minute video call to share our scenario injects and discuss alignment w

Sincerely,

Krishnakumar Radhakrishnan
Lead PI & Architect, HSRI Research Consortium
krishnakumar.rmenon@gmail.com | https://humanreadinessindex.org/
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Pitch Email #2: ETH Zurich Center for Security Studies (CSS)

Subject: Sovereign Decoupling Benchmark: Tabletop Pilot Proposal with ETH Zurich CSS (Risk & Resilience)

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Dear Dr. [Name] / CSS Risk & Resilience Team,

I am writing from the Human Superintelligence Readiness Institute (HSRI Research Consortium).

Our open-source research infrastructure (DOI: 10.5281/zenodo.10842000) recently formalized the Civ

In our comparative benchmark across 39 nations, Switzerland exemplifies "Archetype C (Decentralize

We propose partnering with ETH Zurich CSS to host a 4-hour facilitated tabletop exercise ("Operati

The host financial cost is zero (funded under our research grant). ETH Zurich CSS will co-author H

The full protocol and interactive simulation are live at: https://humanreadinessindex.org/research

Would you be open to a 15-minute introductory call to review the exercise injects?

Warm regards,

Krishnakumar Radhakrishnan
Lead PI, HSRI Research Consortium
krishnakumar.rmenon@gmail.com | https://humanreadinessindex.org/
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8. Pilot Execution Timeline (21-Day Sequence)

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Day 1 - 5   | Initial outreach to RSIS Singapore & ETH Zurich CSS leadership via Pitch Emails.
Day 6 - 10  | 15-minute briefing calls; walk through simulation engine and crisis injects.
Day 11 - 14 | Execution of 1-page MOU; confirm venue and participant attendee list (12-16 seats).
Day 15 - 20 | Distribute confidential participant packets (threat background & role descriptions).
Day 21      | Facilitate 4-hour tabletop exercise under Chatham House Rule.
Day 22 - 30 | Synthesize empirical findings; author and co-release HSRI Case Study 01 on Zenodo.
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